

Abstract

With the increasing number of vehicles in urban areas, traffic congestion (private and public vehicles) and pedestrian safety, especially for elderly, children and special users, have become concerning challenges for modern transportation systems. Typical traffic signal systems are generally based on fixed-time scheduling, where signals operate according to predefined durations regardless of real-time traffic conditions and designed to focus mostly on motorized vehicle movements. This can increase pedestrian waiting times, reduce accessibility, and create safety risks at busy intersections.

The proposed solution suggests design and implementation of an adaptive traffic signal system through software simulation for intersections with vehicle lanes and pedestrian crossings. The implementation is developed using Python and combines a console-based simulation with graphical visualization. The project applies core computer science concepts such as queue data structures, state-based modelling, simulation, control logic, and algorithm design. Mathematical evaluation is included to assess system performance through measurable metrics such as average waiting time, traffic throughput, and fairness ratio between pedestrians and vehicles. This project addresses this issue by designing a traffic signal system that dynamically adjusts according to the motorized traffic flow and pedestrians crossing requests.